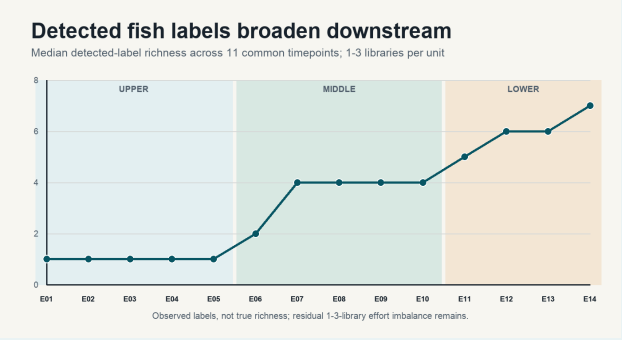
More data, less certainty? A River Conwy eDNA reanalysis

Community ecology | imperfect detection | reference-database currency

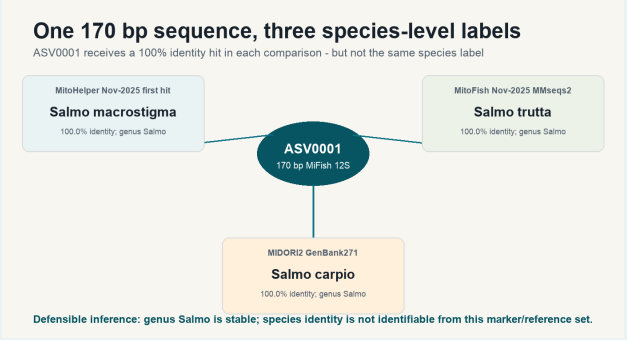
1 River pattern



Across 11 common timepoints, detected-label richness broadens downstream with residual 1-3-library effort; the restricted test is p=0.0714.

NON-CONCLUSIVE - 64/72 BELOW 90%

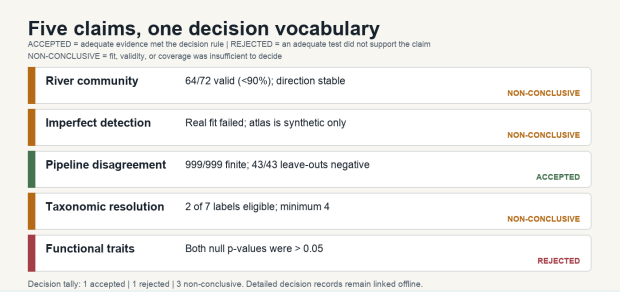
2 Molecular detective



The same 170 bp ASV receives three 100%-identity species labels; only genus is stable.

NON-CONCLUSIVE - SPECIES COVERAGE

3 Claim stress test



One claim is accepted, one is rejected, and three are non-conclusive; every decision includes its reason.

1 ACCEPTED | 1 REJECTED | 3 NON-CONCLUSIVE

Central lesson: rejected and non-conclusive outcomes remain scientific results when their reasons stay visible.

Local portfolio edition. Rare-species occupancy magnitudes remain exploratory; no production MCMC, publication, or hosted service is required.

Source dataset: Perry et al. (2024), Nature Communications 15:4372. doi:10.1038/s41467-024-48640-3